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Laboratory Report – 7

Objective

Generating BEM and nDSM and understanding these data products.

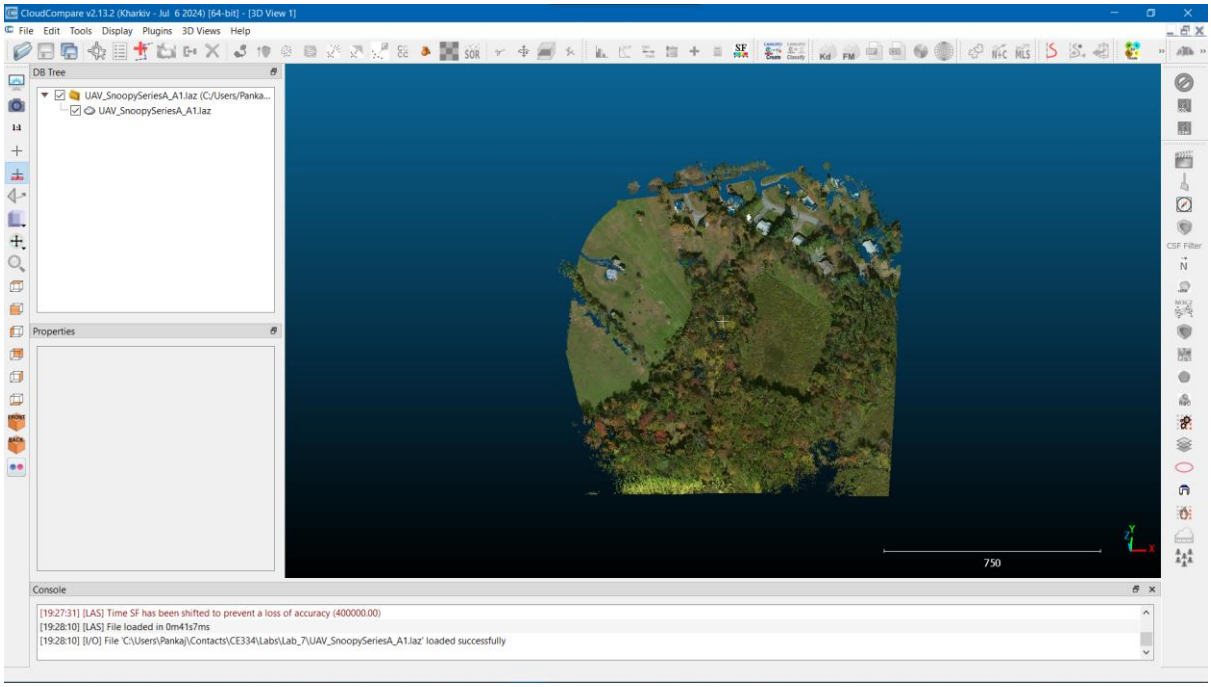
Report

Report should contain the following:

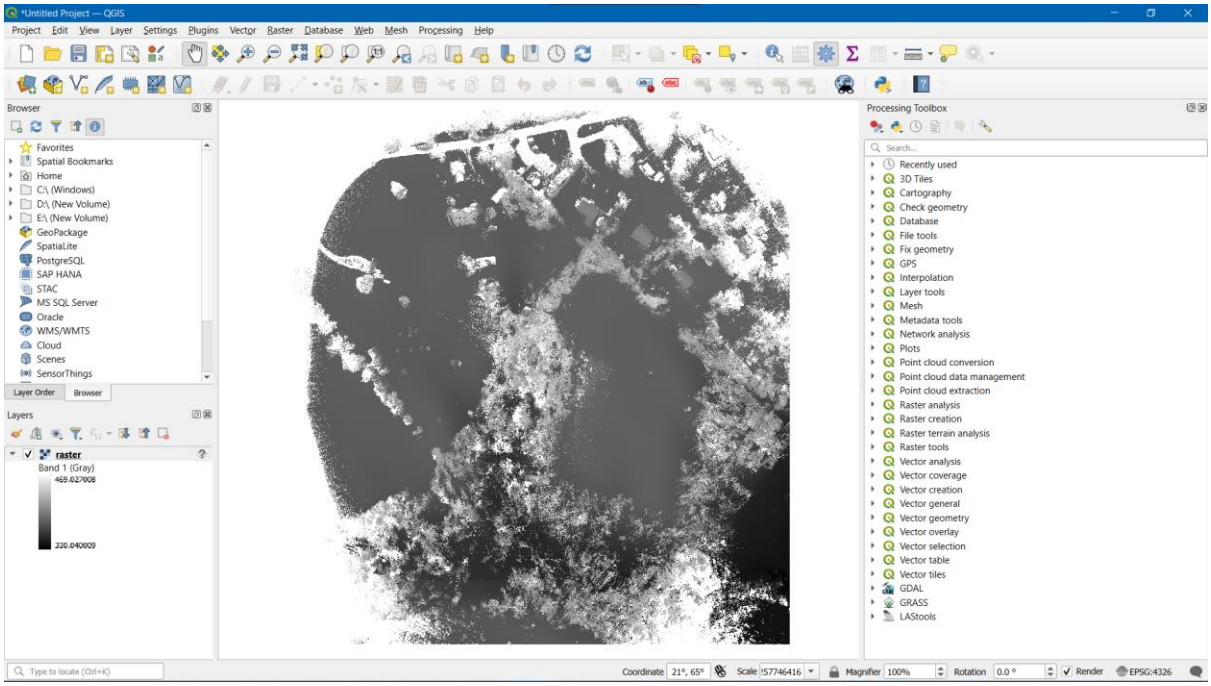
- Images of original data.
- DSM and nDSM images created using the same vertical scale. Try to understand why we want the same vertical scale for these two data sets and write in report.
- Observation on this method. Also, the comment on how BEM and nDSM can be generated by directly working with point cloud and not converting these to raster.
- Also, write in what kind of terrain this method will fail.

Methodology with Q-GIS and CloudCompare

- We had used the same file employed in the last exercise i.e. UAV_SnoopySeriesA_A1.laz
- Observe the point data in a viewer. In this case we used CloudCompare to view the data points.

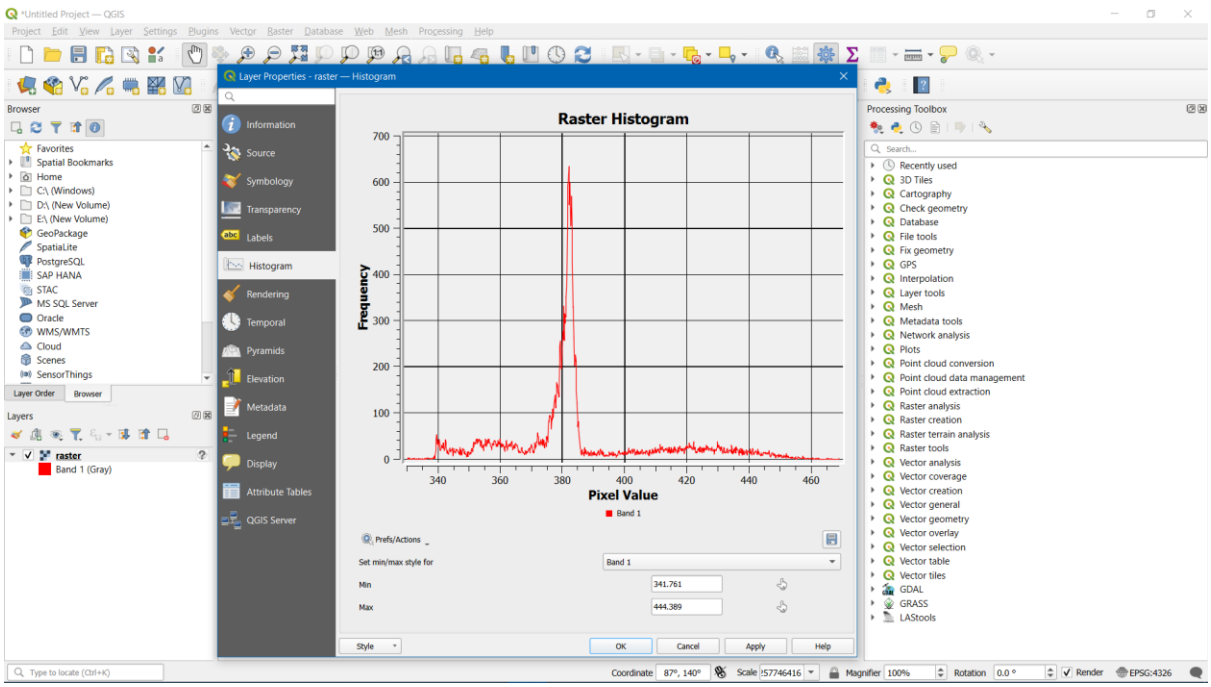


- We then created a raster DSM of raw data where cell values are elevation values. Use the steps that are used to generate DSM in the previous lab.



➢ We generate a histogram of data i.e. frequency vs height of point by using following steps. Observe the histogram.

- Load the DSM layer (.tif file) into QGIS.
- In the Layers panel located on leftside, you can see your loaded layer.
- Right click on the layer and select properties.
- Switch to histogram tab and compute the histogram.



➢ Decide a threshold on the basis of the raw data, raster image and histogram displays which can separate above ground objects from the ground.

- Specify the threshold = 386 and write reason for selecting the threshold.

Reasoning: In a typical LiDAR dataset, the histogram will show a large peak representing the ground. The threshold should be set just above this ground peak to separate terrain from "above-ground" objects like buildings and trees.

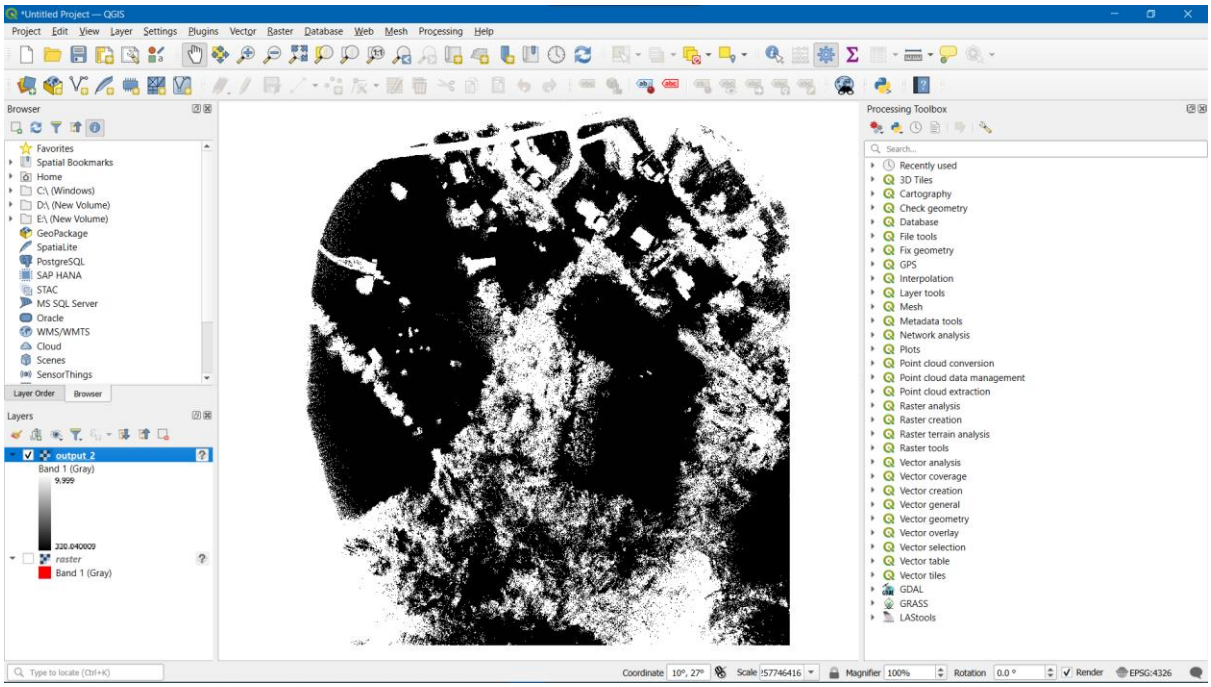
- Apply the threshold on the DSM image by converting above ground pixels as 9999 and the rest with their original values.

  a. Go to the Raster Tab in the menu toolbar and select the raster calculator.

  b. Type the expression: if("raster@1" > 386,9999,"raster@1")

  c. Specify the output file path and its name.

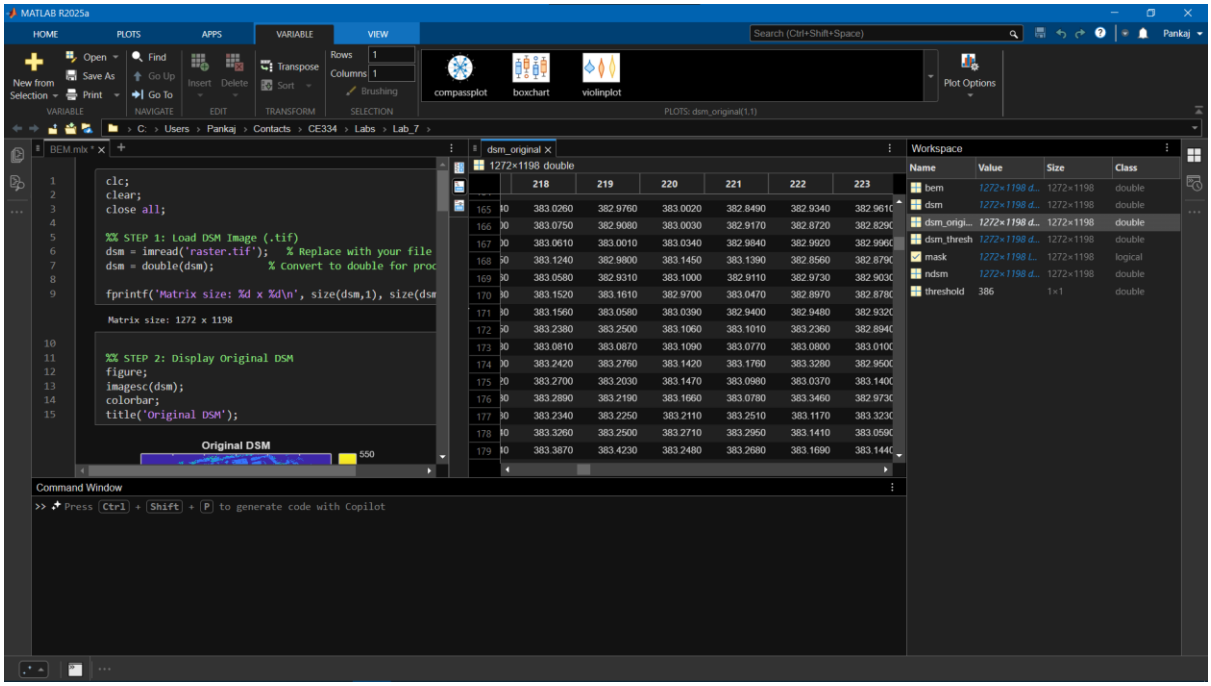
  d. Hit OK.



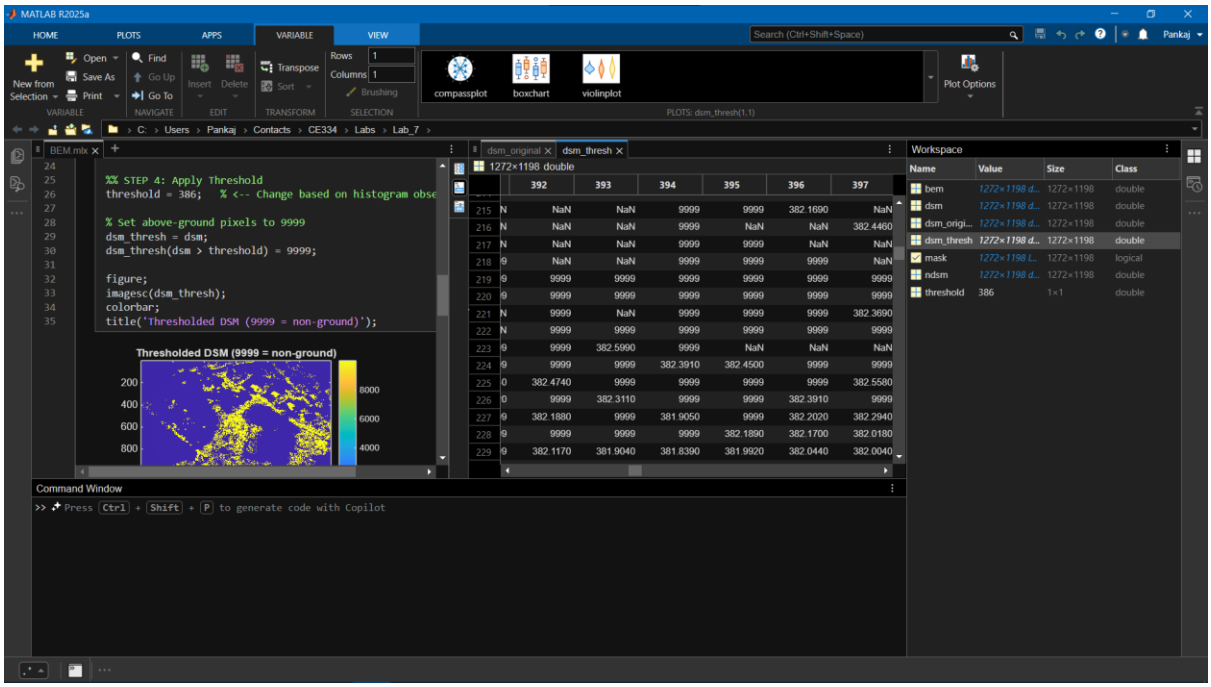
White: Represents pixels with the value **9999** (the highest values in the dataset), which are the above-ground objects you masked.

Black/Darker: Represents the original **ground elevation** values, which are significantly lower than 9999.

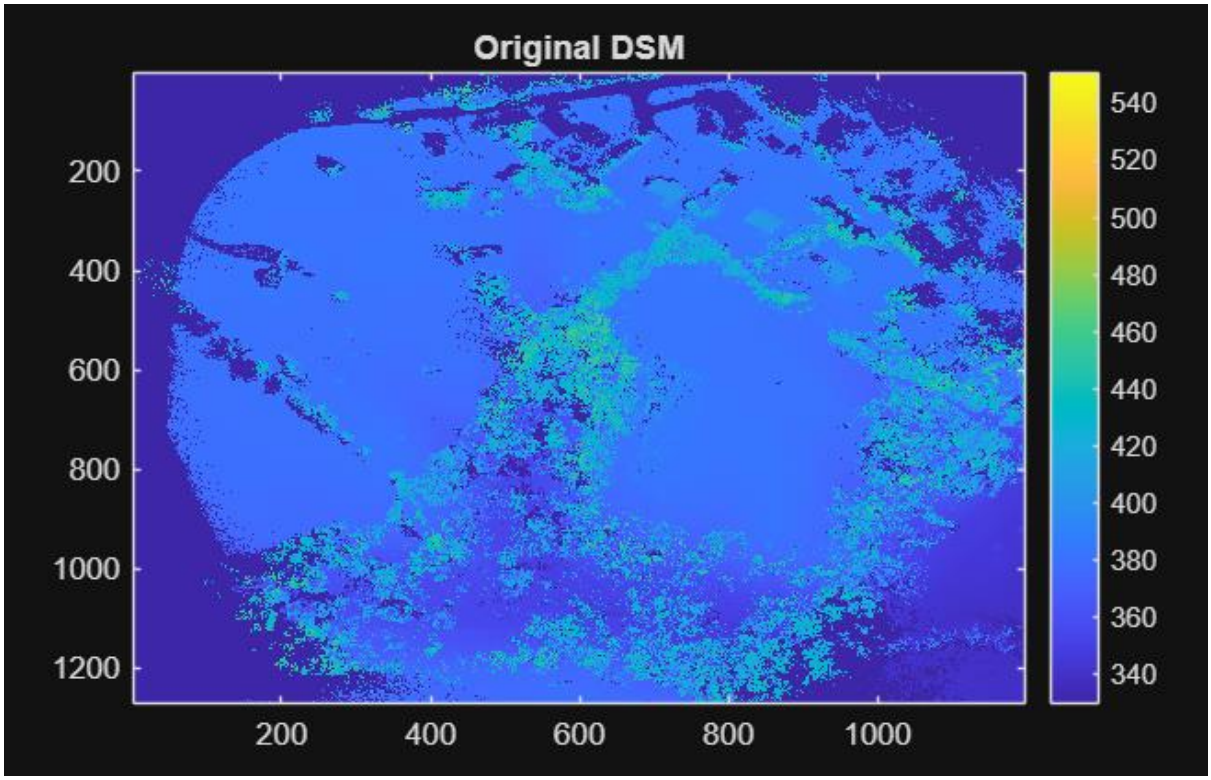
- Try to do the steps (a) to (d) by loading the Raster file in MATLAB.
- Remove each 9999 pixels by interpolation from neighbouring pixels (i.e. ground pixels).
  - Load the .tif file generated after changing the pixel values into MATLAB using imread function.
  - You can see that your data is stored in the form of matrix.



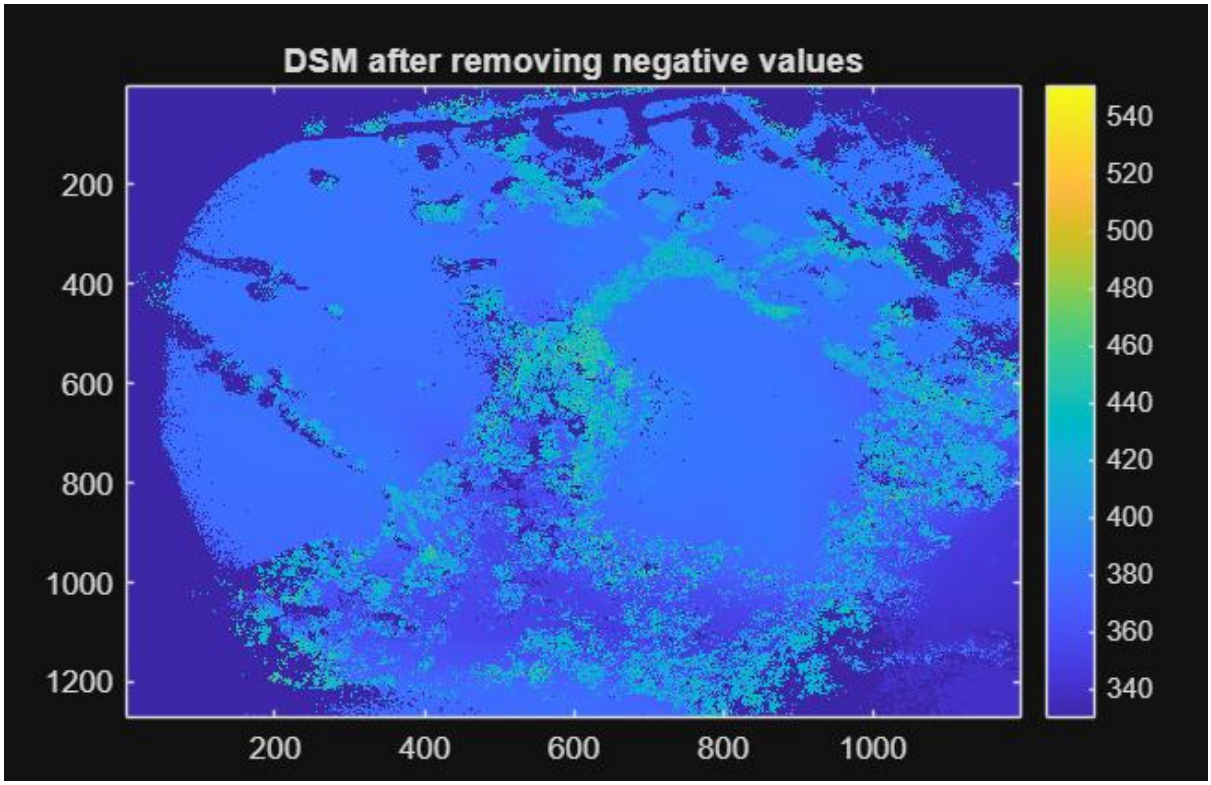
- Write the dimensions of the matrix i.e. **Matrix size: 1272 x 1198**
- Open the variable in which the image is stored and look on its values. You can see that some cells have a value 9999 which you have changed earlier.



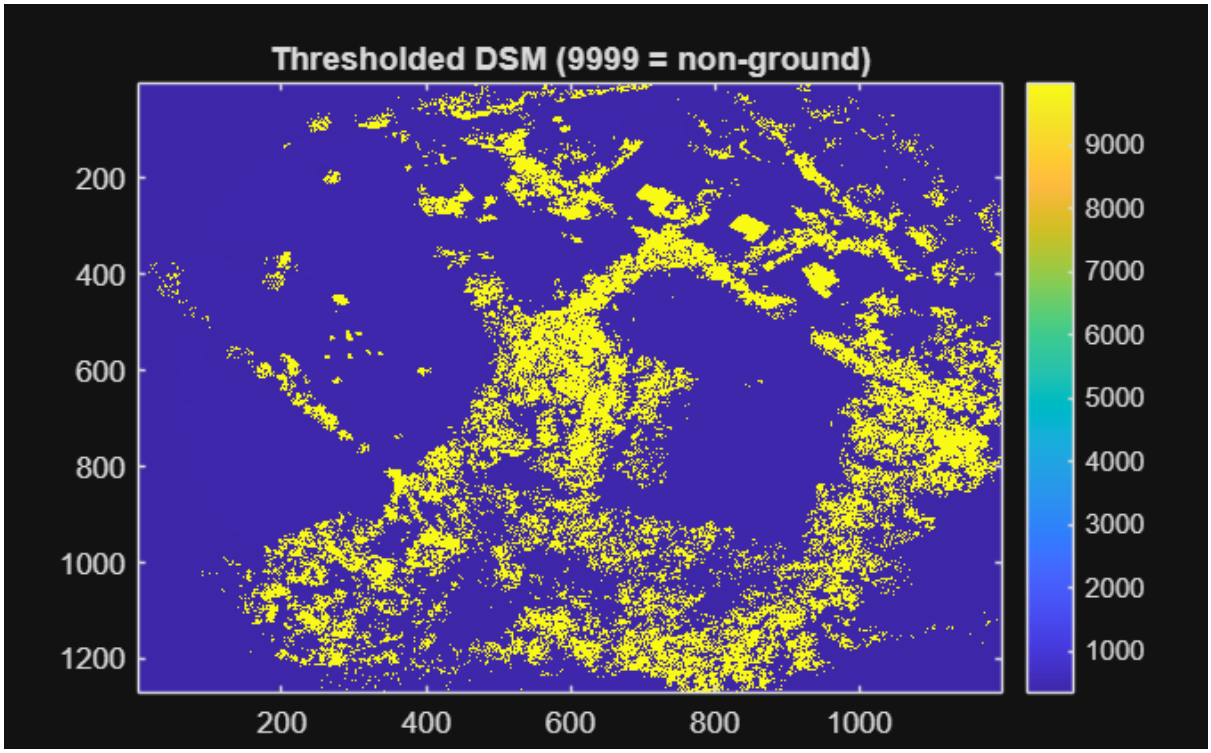
- You can also see some cells which have a large negative value.
- Display the loaded image using “imagesc(variable)”.



- Then try to change the values which are less than 0 in the matrix to zero.

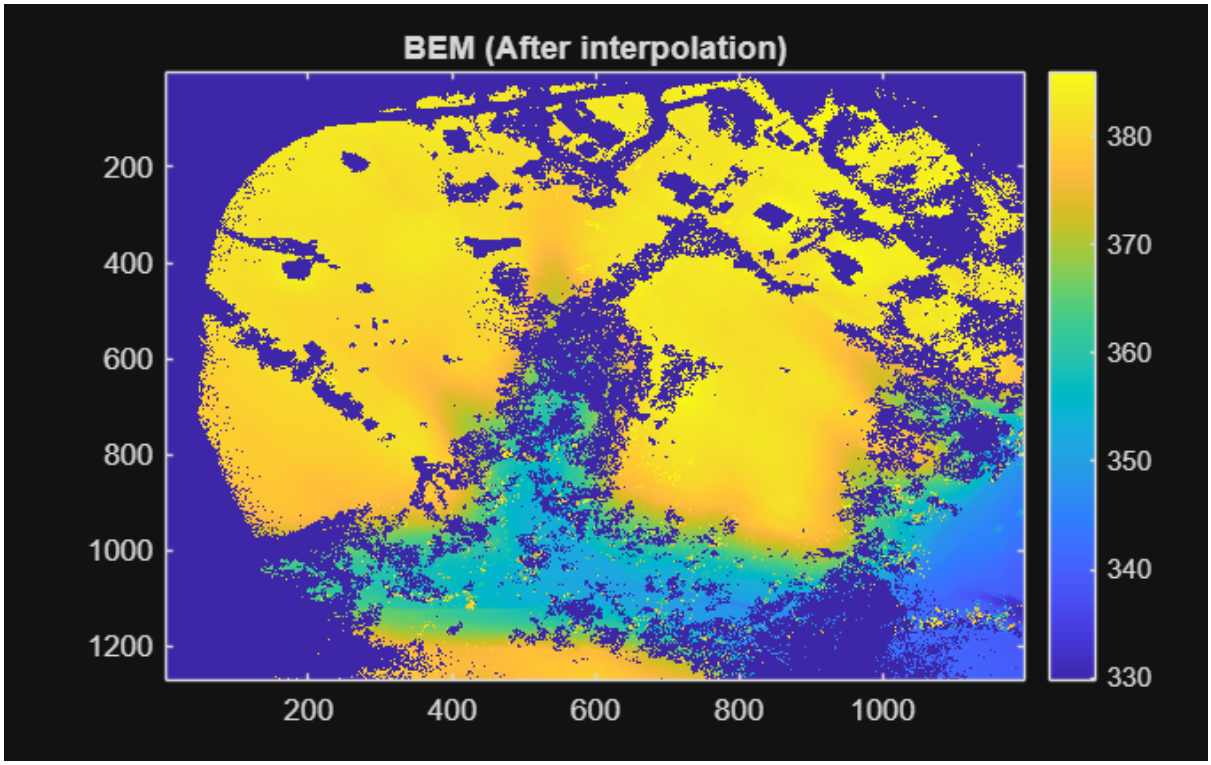


- Write the changes you observed in the image and attach the loaded image before and after changing the values in the report.
  By setting values < 0 to 0, the image becomes cleaner in imagesc because the color scale is no longer stretched by extreme negative outliers.
- Then create a mask of the pixel values which are equal to 9999 (remember cells in the matrix represent the pixels of your image.)



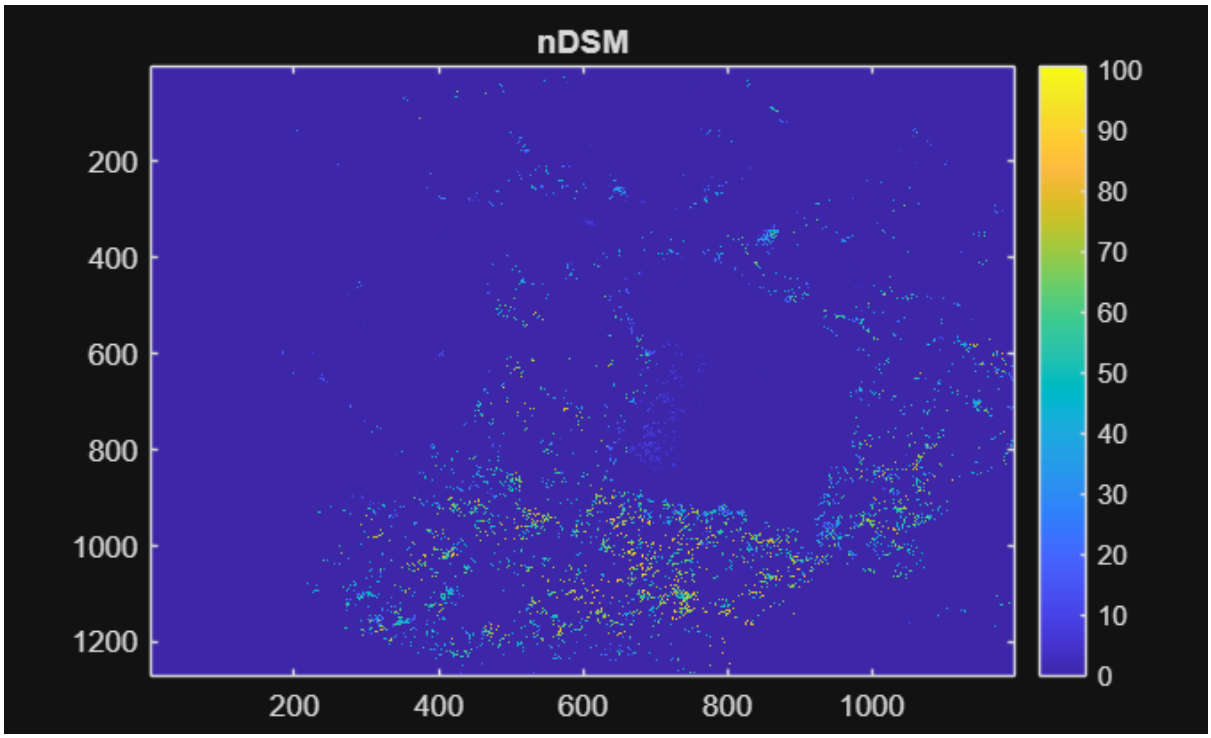
- Then, Use the region fill command to remove the pixels with 9999 values and fill it with interpolated values.

➢ This is the BEM.



Alternative Methods: Other ways to interpolate include **Kriging**, **Inverse Distance Weighting (IDW)**, or **Nearest Neighbour** interpolation

- Open the variable in which you stored the interpolated values and try to compare the height values of the non-ground pixels.
    - Load the DSM file in the matlab.
    - Find nDSM = DSM – BEM (Since both the variables are matrices, so basic matrix operations apply here).
    - Display nDSM in matlab image viewer and check the pixel values.



- The ground pixel values after subtraction will be **zero**
  Why is this image called as nDSM.
  It is called a Normalized Digital Surface Model (nDSM) because it "normalizes" the terrain to a flat baseline of zero, leaving only the actual heights of objects (buildings, trees) relative to the ground.

Answer

Why use the same vertical scale?

- Using the same scale allows for a direct visual comparison. If the scales differ, a 10-meter building in the nDSM might look taller or shorter than the same building in the DSM, making it difficult to verify the accuracy of the subtraction.

Point Cloud vs. Raster

- BEM and nDSM can be generated directly from point clouds by classifying points as "Ground" or "Non-Ground" using algorithms like **Cloth Simulation Filter (CSF)**. This avoids the data loss that occurs during rasterization.

Terrain Failure Cases

- This specific method (simple thresholding and interpolation) will likely fail in **steep, mountainous terrain**. In such areas, the ground elevation changes so rapidly that a single threshold might classify a high-altitude ground point as an "object" or a low-altitude building as "ground".